

EDUCATION

- **University of Ottawa** Ottawa, ON
Joint Honours BSc in Computer Science and Mathematics; GPA: 3.8/4 *Sep. 2023 – Apr. 2028*

PROGRAMMING SKILLS

- **Languages:** Java, Python, Python, SQL, R, C/C++, JavaScript/TypeScript
- **Technologies:** Git, Home Server, AWS, Docker, Spring, PostgreSQL, Postman, SonarQube, Jira, WebSphere, Openshift

EXPERIENCE

- **Innovation, Science and Economic Development Canada** Ottawa, ON
API Software Developer Intern, Patent Branch *Jan 2026 - Present*
Tech Stack: Java, Spring Boot, REST, OpenShift, API, SQL, Jenkins, XML
 - Developed and maintained various Java-based APIs and backend services in the Canadian Patent Filing System and Canadian Public Database, supporting millions of patents records and associated legal documents.
 - Debugged and resolved a concurrency issue in an asynchronous document-processing pipeline, tracing overlapping requests and temporary-file cleanup to prevent premature deletion of documents.
 - Maintained Apache Camel document submission routes, implementing asynchronous processing, Salesforce and FileNet integration while handling the lifecycle of legal documents.
 - Worked on WIPO ST.96 compliant XML generation for patent publication data, extracting data using SQL/database functions while navigating XSDs, XSLTs, and business specific schema extensions.
 - Troubleshooted WebSphere application-server setup and deployment issues, analyzing configurations and documenting solutions for the team and future developers.

PROJECTS

- **Home Sever & API Gateway** June 2026 - Present
Building with: Docker, Go, Tailscale, Jenkins, Portainer, WSL2
 - Built and configured a self-hosted server from spare PC components to host projects
 - Designed and implemented a Go-based API gateway for routing requests across self-hosted containerized services.
 - Implemented path-based service routing, CORS policies, and a centralized token bucket rate limiting to control and secure access to backend services.
 - Built Jenkins CI/CD pipelines to automate application builds and deployments, with Docker containerization and Portainer for service management and monitoring.
 - Configured Tailscale funnel and networking to securely expose and connect services without directly exposing individual backend applications.
- **Remote Receipt Printer API** Feb 2026 - Present
Building with: Python, FastAPI, REST APIs, and Tailscale on self-hosted infrastructure
 - Deployed a publicly accessible RESTful API using Python FastAPI that enables remote printing of images and text onto a local printer by exposing the self-hosted service securely through Tailscale Funnel.
 - Implemented idempotency UUID validation to secure the API from abuse and duplicate submissions.
 - Tested API endpoints using Postman, validating request/response handling and error scenarios before deployment.
 - Implemented asynchronous job processing with local printer queue, decoupling print submissions from printer availability during offline periods.
- **Bus Tracking System** Jan 2025 - Present
Building with; Spring Boot, Hibernate, PostgreSQL, Docker, PostGIS
 - Developed a Spring Boot REST API that processes GTFS and GTFS-Realtime data from multiple transit agencies to provide live vehicle locations and transit service information.
 - Designed a PostgreSQL/PostGIS data model for storing and querying large transit datasets, implementing database indexes and geospatial queries to optimize data retrieval.
 - Implemented real-time vehicle processing by correlating vehicle positions with trip and stop-time data to identify active vehicles and calculate route-level performance metrics.
 - Fully containerized application and database for reproducible and simplified deployments.